

ReSNA: A Robotic-Enabled Surgical Navigation Assistant for Breast-Conserving Surgery

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Abstract—Breast-conserving surgery (BCS) is a standard treatment for early-stage breast cancer; however, accurate tumor localization during surgery remains challenging due to significant breast morphological differences between preoperative diagnostic images and intraoperative presentation in conjunction with limited intraoperative imaging information. The unfortunate result is high reoperation rates. To address this limitation, unlike most existing systems that employ robots primarily as actuators, a robotic-enabled surgical navigation assistant (*ReSNA*) is realized that integrates a novel “eyes-in-hand” dual-vision robotic platform with active robotic surveillance and AI-driven surface tracking as an automated observer to acquire intraoperative breast surface measurements. These measurements drive a non-rigid image-to-physical registration to align preoperative diagnostic imaging data to the deformed breast intraoperatively and can subsequently be used to improve tumor localization and margin navigation. *ReSNA* is initially validated on a patient-specific breast phantom with embedded targets under multiple deformation states and achieved an average target registration error (TRE) of 3.4 ± 1.4 mm, significantly outperforming rigid registration (11.3 ± 7.0 mm), showing clear potential in advancing clinical capability for image-guided BCS.

Index Terms—Surgical robotics: planning, medical robots and systems, computer vision for medical robotics, surveillance robotic systems.

I. INTRODUCTION

BREAST-CONSERVING surgery (BCS), commonly referred to as lumpectomy or partial mastectomy, is a standard treatment for many patients with early-stage breast cancer because, when combined with appropriate adjuvant therapy, it

preserves breast tissue while achieving similar outcomes to total mastectomy. Unfortunately, across treatment centers, 20-40% of BCS patients can experience additional operations, depending on tumor type, margin criteria, and surgical practice [1]. One contributor to these high re-excision and re-operation rates is the difficulty in localizing and controlling tumor margin navigation during surgery. For BCS, lesion spatial extent is typically assessed from preoperative imaging data and localized with a seed or guidewire prior to surgery. Intraoperatively, these are used to provide a path to the lesion location but do not provide any sense of lesion boundary. Although intraoperative ultrasound may be employed, only 50% of nonpalpable tumors are visible in the breast using ultrasound [2]. With respect to available preoperative imaging data, standard diagnostic imaging involves mammography with the breast compressed, while MRI is typically performed with the patient lying prone and the breast pendant. Both imaging positions differ from the surgical configuration, where the patient lies supine with the ipsilateral arm extended 90° [3]. The differences among states cause the tumor to undergo large shifts and deformations when presented intraoperatively [4]. Supine MR imaging has been proposed as a solution [5], but, even in this case, significant deformations have been shown to persist due to arm positioning differences [6]. To mitigate these challenges, non-rigid (deformable) registration can be employed to account for breast deformation. For instance, several studies have used finite element methods (FEM) or biomechanics-based methods to model and compensate for breast deformation [7], [8]. While these methods have shown promise, accurately acquiring sufficient intraoperative data to drive these models without disrupting surgical workflow remains an active research challenge. Mazier et al. [9] and Ruiz et al. [10] utilized mobile 3D scanners for intraoperative breast surface acquisition, but these methods required complex manual interventions that significantly interfered with the surgical workflow. Richey et al. [11] introduced a dual-vision system using a passive mechanical arm to surveil breast shape changes. While this work was encouraging, it lacks adaptability and the digitization workflow was cumbersome.

Collectively, these experiences highlight the need for a streamlined solution capable of acquiring sufficient intraoperative data for assisting navigation in the presence of deformation while preserving the surgical workflow. To address, our novel platform *ReSNA* is introduced here and comprises the following characteristics: (1) a novel, lightweight, and modular “eyes-in-hand” non-contact robotic system that observes the surgical

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Vanderbilt University Institutional Review Board under Application No. 130038, and performed in line with the Declaration of Helsinki.

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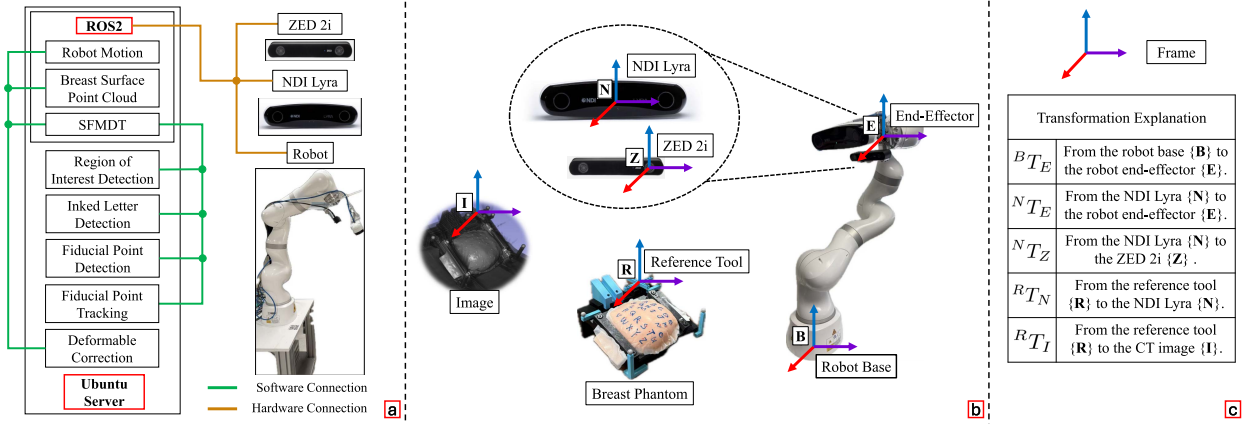


Fig. 1. *ReSNA* System Overview. (a) Programming interface: the robot motion is controlled by ROS 2. The surface fiducial marker detection and tracking (SFMDT) and deformable correction pipelines are running on a workstation; (b) System components: the dual-vision asset is mounted on a KUKA robot to track the breast surface field and the reference tool $\{R\}$, which connects the patient, robot, and imaging space; (c) Context explanation: the transformations used in this study.

scene and measures breast surface data and tool locations for use in subsurface tumor localization; (2) an automated intraoperative data surveillance strategy enabled by a new learning-based vision pipeline and active robot-assisted motion; and (3) an enhanced deformable correction pipeline that accurately aligns tumor location and other features of interest from preoperative images to the intraoperative presentation.

II. METHODOLOGY

The LBR Med 14 R820 robot (KUKA, Germany) arm was utilized for this study, as it is certified for integration into medical devices which enables more efficient pathways for FDA clearance. An optical tracking sensor, Polaris Lyra (Northern Digital Inc., Waterloo, ON, Canada) was used to provide accurate tracking of the reference tool (shown in Fig. 1), and a ZED 2i stereo camera (Stereolabs Inc., San Francisco, CA, USA) was used to acquire scene images and composite 3D textured point clouds.

A. System Overview and Calibration

Fig. 1 shows an overview of the *ReSNA* system. The dual-vision “eyes” consist of the Lyra and ZED 2i sensors mounted on the robot end-effector using a custom aluminum-alloy bracket. The bracket positions and tilt angles were carefully designed to provide overlapping fields of view. The robot base is fixed on a custom movable cart, and the complete dual-vision assembly is lightweight, with a total mass of less than 3 kg. Lyra’s continuous tracking of a reference tool [12] $\{R\}$ provides a shared spatial anchor that registers preoperative imaging data to the patient body, with its placement choice detailed in Section II-C1. Hand-eye calibration using the least squares method between the Lyra and the robot end-effector was performed by fixing a tracking tool (No. 8700339, Northern Digital Inc., Waterloo, ON, Canada) within the Lyra’s field of view and moving the end-effector to ten different positions and orientations. The hand-eye transformation $\mathbf{X} = {}^N T_E^{-1} \in SE(3)$ satisfies the classical $\mathbf{A}_i \mathbf{X} = \mathbf{X} \mathbf{B}_i$ formulation [13], where $\mathbf{A}_i$ are relative motions

of the robot, and $\mathbf{B}_i$ are relative motions of the Lyra. Next, calibration between the Lyra and the ZED 2i was performed using a 7×8 checkerboard equipped with four retro-reflective passive sphere markers (Northern Digital Inc., Waterloo, ON, Canada) [14]. The calibration was conducted by positioning the checkerboard and teleoperating the robot in different configurations to collect ten sets of data, resulting in a total of 420 corner correspondences.

We used ROS 2 Humble [15], [16] with the Orocos KDL library [17] for motion control and path planning, acquiring Lyra and ZED 2i data via the C++ Combined API (Northern Digital Inc., Waterloo, ON, Canada) and the ZED SDK ROS 2 wrapper (Stereolabs Inc., San Francisco, CA, USA).

B. Intraoperative Data Acquisition

ReSNA acquires two types of intraoperative breast data: surface fiducial markers and a breast surface point cloud. We utilized the surface fiducial marker design from a prior study [18], which comprises a fiducial point, represented as a pixel position $\mathbf{I}_j$ in 2D or a 3D position $\mathbf{P}_j$ in space, and an inked letter label $\ell_j \in \{A, \dots, Z\}$ encoding its correspondence. Together, a set of N markers is denoted as $\{(\mathbf{x}_j, \ell_j)\}_{j=1}^N$, where $\mathbf{x}_j$ refers to either the 2D pixel position $\mathbf{I}_j$ or the 3D spatial position $\mathbf{P}_j$ depending on context. Their use will be discussed in Section III-B.

1) *Surface Fiducial Marker Detection and Tracking*: To efficiently identify the surface fiducial markers without interfering with the surgical workflow, we developed a novel acquisition approach that overcomes the semi-manual processes proposed in [18] by employing a four-stage vision-based machine-learning pipeline (Fig. 2(a)) to obtain, for each image frame t , a set of $N^{(t)}$ detections $\{(\mathbf{I}_j^{(t)}, \ell_j^{(t)})\}_{j=1}^{N^{(t)}}$. The dataset contained 419 images (343 breast, 76 phantom). Breast images were extracted from surveillance videos of seven volunteers as approved by the Institutional Review Board at Vanderbilt University (date of approval 11/11/2015) that captured realistic deformations caused by arm position changes and field-of-view occlusions

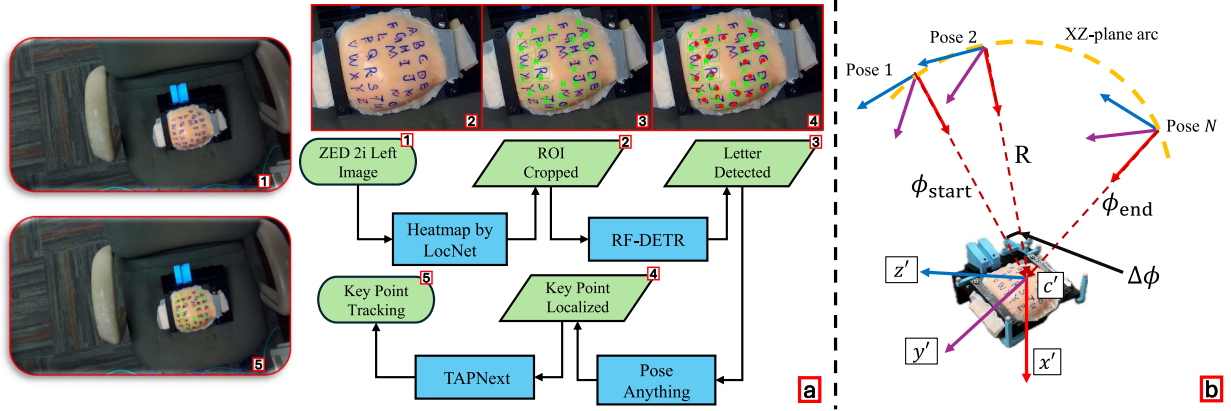


Fig. 2. Intraoperative data acquisition. (a) Surface fiducial marker detection and tracking pipeline: a demo image is used to illustrate the pipeline workflow; (b) Robot Motion Assistant: the XZ-plane arc of the *curved-cross* trajectory.

during ultrasound image acquisition. Phantom images were acquired from multiple viewpoints of a physical phantom (different than the one in Section III). Breast images from four volunteers (217) were used for training, and from the other three volunteers for validation¹ (64) and testing (62). The 217 training images were further augmented tenfold with different data augmentation methods. All images were annotated with bounding boxes across 26 surface fiducial marker categories. The four-stage process involved:

Stage One, Region of Interest Detection: To identify the breast region in the image, we proposed LocNet (see appendix for details). We used both breast and phantom images in the training set. The bounding-box annotations were converted into binary masks serving as ground-truth heatmaps. At inference, high-activation regions were automatically bounded as the breast region.

Stage Two, Inked Letter Detection: We fine-tuned a pre-trained *RF-DETR-Small* model [19] to detect inked letters associated with fiducial points on the breast surface. The dataset was identical to stage one, but was pre-processed through the trained LocNet to crop interested breast regions.

Stage Three, Fiducial Point Detection: To localize the fiducial point associated with each detected inked letter, whose position may vary arbitrarily relative to the letter's appearance, we trained (see appendix for details) a *Pose Anything* model [20] in the 5-shot configuration.

Stage Four, Fiducial Point Tracking: We employed the TAPNext [21] without re-training to track each fiducial point detected from the previous stage.

This novel combination of network models combined with our surface fiducial marker approach yielded a robust surveillance framework for soft-tissue measurement in breast.

2) **Robot Motion Assistant:** To enable robotic assistant in intraoperative data acquisition, we design a *curved-cross* trajectory, allowing active adjustment of the “eyes” across multiple viewpoints with two aims: (1) multi-view consensus for surface fiducial marker detection, reducing false positives and recovering missed markers from single-view predictions

caused by variations in lighting, viewing angle, surface reflections, and partial occlusions; and (2) multi-view surface point cloud acquisition, where observations from different viewpoints are fused using the robot kinematics to reconstruct a comprehensive breast surface that captures the complete breast geometry.

The *curved-cross* trajectory consists of a set of position- and orientation-paired poses $\{(\mathbf{p}_{xz,k}, \mathbf{q}_{xz,k}), (\mathbf{p}_{xy,k}, \mathbf{q}_{xy,k})\}$ on two perpendicular arcs (the XZ-plane arc and the XY-plane arc in Fig. 2(b)) on a sphere with radius R and center at $\mathbf{c}'$ on breast. The position of $\mathbf{c}'$ is obtained from the Lyra-tracked reference tool center $\mathbf{c} = [x_c, y_c, z_c]^\top$ with an offset $\delta = [\delta_x, \delta_y, \delta_z]^\top$ to breast, yielding $\mathbf{c}' = \mathbf{c} + \delta$. To compute $\mathbf{p}_{xz,k}$, the angular range $\phi_{start} \rightarrow \phi_{end}$ is discretized with step size $\Delta\phi$, resulting in sampled angles $\phi_k = \phi_{start} - k\Delta\phi$ and positions $\mathbf{p}_{xz,k} = [x'_c + R \cos \phi_k, y'_c, z'_c + R \sin \phi_k]^\top$. For $\mathbf{q}_{xz,k}$, we first obtain $\mathbf{q}_{xz,1}$ and $\mathbf{q}_{xz,N_{xz}}$, which orient the end-effector toward $\mathbf{c}'$. The intermediate orientations (from $\mathbf{q}_{xz,2}$ to $\mathbf{q}_{xz,N_{xz}-1}$) are then generated using spherical linear interpolation (slerp) [22] as $\mathbf{q}_{xz,k} = \text{slerp}(\mathbf{q}_{xz,1}, \mathbf{q}_{xz,N_{xz}}, t_k)$, where $t_k = \frac{k}{N_{xz}}$. Similarly, for $\mathbf{p}_{xy,k}$, the XY-plane angles are sampled as $\theta_k = \theta_{start} - k\Delta\theta$, with positions $\mathbf{p}_{xy,k} = [x'_c + R \cos \theta_k, y'_c + R \sin \theta_k, z'_c]^\top$. The orientations $\mathbf{q}_{xy,k}$ are obtained in the same manner as $\mathbf{q}_{xz,k}$. By setting the parameter values (see Section III-B) on a breast-specific basis, the trajectory can adapt to different operating environments and patients with varying breast geometries.

As the end effector moves along the *curved-cross* trajectory, the ZED 2i continuously acquires an image and a registered 3D point cloud $\mathbf{P}_{\text{intra-surf}_z}^{(t)}$ per frame t . Each image is streamed to the server and processed by the surface fiducial marker detection and tracking pipeline in Section II-B1, yielding per-frame detections $\{(\mathbf{I}_j^{(t)}, \ell_j^{(t)})\}_{j=1}^{N^{(t)}}$. Each $\mathbf{I}_j^{(t)}$ is then unprojected onto $\mathbf{P}_{\text{intra-surf}_z}^{(t)}$ to obtain the corresponding 3D fiducial position $\mathbf{P}_{\text{intra-fid}_z,j}^{(t)}$, which is subsequently transformed into the robot base space as $\mathbf{P}_{\text{intra-fid}_b,j}^{(t)}$ by using ZT_B , where ${}^ZT_B = {}^NT_Z^{-1} \cdot {}^NT_E \cdot {}^BT_E^{-1}$.

The per-frame detections $\{(\mathbf{P}_{\text{intra-fid}_b,j}^{(t)}, \ell_j^{(t)})\}_{j=1}^{N^{(t)}}$ are then consolidated into a single set of intraoperative surface fiducial markers $\mathbf{P}_{\text{intra-fid}_b}$ via Algorithm 1. Then, ZT_B is also applied

¹Not used for hyperparameter tuning or backpropagation.

Algorithm 1: Cross-frame Fiducial Marker Consensus.

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1: Input:  $\{(\mathbf{P}_{\text{intra-fid\_b},j}^{(t)}, \ell_j^{(t)})\}_{j=1}^{N^{(t)}}$  of each frame  $t$ ,
    $\epsilon = 5$  mm
2: Output: A set of intraoperative surface fiducial markers:
    $\{(\mathbf{P}_{\text{intra-fid\_b},j}, \ell_j)\}_{\ell_j \in \{A, \dots, Z\}}$  (denoted as  $\mathbf{P}_{\text{intra-fid\_b}}$ )
3: Note:  $\text{GROUP}(\cdot, \epsilon)$  returns groups of detections whose
   points fall within a region of radius  $\epsilon$ .
4: function FiducialConsensusInput
5:    $\mathcal{D} \leftarrow \{(\mathbf{p}, \ell_j^{(t)}) \mid \mathbf{p} \in \mathbf{P}_{\text{intra-fid\_b},j}^{(t)}, \forall j, t\}$ 
6:   for each unique letter  $\ell_j \in \{A, \dots, Z\}$  do
7:      $\mathcal{D}_j \leftarrow \{(\mathbf{p}, \ell) \in \mathcal{D} \mid \ell = \ell_j\}$   $\triangleright$  Filter detections for
        $\ell_j$ 
8:      $N_{\ell_j} \leftarrow |\mathcal{D}_j|$   $\triangleright$  Total detection count
9:      $\mathcal{C}_j \leftarrow \text{GROUP}(\mathcal{D}_j, \epsilon)$   $\triangleright$  Group detections within  $\epsilon$ 
10:     $\mathcal{C}_j^* \leftarrow \{C \in \mathcal{C}_j \mid |C| \geq \lceil \frac{1}{2} N_{\ell_j} \rceil\}$   $\triangleright$  Retain valid
      groups
11:     $C^* \leftarrow \arg \max_{C \in \mathcal{C}_j^*} |C|$   $\triangleright$  Select largest valid group
12:     $\mathbf{P}_{\text{intra-fid\_b}}(\ell_j) \leftarrow \frac{1}{|C^*|} \sum_{\mathbf{p} \in C^*} \mathbf{p}$   $\triangleright$  Compute centroid
13:  end for
14:  return  $\{(\mathbf{P}_{\text{intra-fid\_b},j}, \ell_j)\}_{\ell_j \in \{A, \dots, Z\}}$ 
15: end Function

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to $\mathbf{P}_{\text{intra-surf_z}}^{(t)}$ to obtain $\mathbf{P}_{\text{intra-surf_b}}^{(t)}$. The $\mathbf{P}_{\text{intra-surf_b}}^{(t)}$ acquired at the four edges and center of the *curved-cross* trajectory are aggregated to reconstruct one intraoperative breast surface point cloud:

$$\mathbf{P}_{\text{agg}} = \bigcup_{t=1}^N \mathbf{P}_{\text{intra-surf_b}}^{(t)}. \quad (1)$$

The $\mathbf{P}_{\text{agg}}$ is then decimated to a minimum inter-point distance of η , ranging from 2–10 mm (see Section III-C).

C. Deformable Correction

To address breast shape discrepancies between the preoperative images and the intraoperative presentation, we implemented a deformable correction pipeline that performs an initial alignment using rigid registration, followed by a non-rigid registration for deformable correction.

1) *Rigid Registration:* Prior work [11] used point-to-point rigid registration of surface fiducial markers between preoperative and intraoperative spaces for initial alignment. However, it may be inaccurate when the data are deformed. Therefore, this work establishes the same alignment but with one key difference: leveraging a reference tool adhered to the sternum (see Section III-A), which remains stable under arm-movement-induced breast deformation and allows for the compensation from respiratory shifts. The rigid transformation ${}^R T_I = [\mathbf{R} \mid \mathbf{t}]$ is computed via: $\min_{\mathbf{R}, \mathbf{t}} \sum_{k=1}^N \|\mathbf{P}_{\text{ref_i},k} - (\mathbf{R} \cdot \mathbf{P}_{\text{ref_r},k} + \mathbf{t})\|^2$, with $\mathbf{P}_{\text{ref_i},k}$ and $\mathbf{P}_{\text{ref_r},k}$ denoting the k -th reference tool point in the preoperative image and reference tool space, respectively.

2) *Non-Rigid Registration:* In this study, we employed a deformable correction method for the breast using the same previously published parameter settings [8]. Briefly described,

the deformation state is parameterized by $\beta = [\alpha, \tau, \theta]$, where $\alpha \in \mathbb{R}^{3k}$ are the basis weights for k control points where each control point creates a 3D volumetric Kelvinlet displacement field, τ is the rigid translation, and θ is the rigid rotation.

The displacement field for the full 3D model $\hat{\mathbf{u}}(\mathbf{x} \mid \beta) = R(\theta)(\mathbf{x} - \mathbf{p}_0 + \mathbb{K}_u \alpha) + \tau + \mathbf{p}_0$ is recovered, where $\mathbb{K}_u \in \mathbb{R}^{3m \times 3k}$ is a 3D Kelvinlet displacement response matrix pre-computed from the preoperative model $\mathcal{M}$ consisting of m vertices, $\mathbf{p}_0$ is the centroid of $\mathcal{M}$, and $R(\theta)$ is the rotation matrix defined by the global rigid rotation vector θ . Therefore, each deformed point position $\mathbf{p}_{\text{cor}}$ can be obtained from its original position $\mathbf{p} \in \mathcal{M}$ using β according to $\mathbf{p}_{\text{cor}} = \mathbf{p} + \hat{\mathbf{u}}(\mathbf{p} \mid \beta)$.

The deformation state β is optimized by minimizing the energy function:

$$\Omega(\beta) = \frac{1}{n_p} \sum_{i=1}^{n_p} (e_p^i)^2 + \frac{1}{n_s} \sum_{i=1}^{n_s} (e_s^i)^2 + w_{\text{SE}} e_{\text{SE}}^2, \quad (2)$$

where the first term, based on the point correspondence error e_p^i for n_p surface fiducial markers, and the second term, based on the point-to-surface distance error e_s^i for n_s non-corresponding surface points sampled from the surface point cloud, represent the data fidelity terms. The final term, e_{SE} , is a strain energy regularization term that penalizes large distorting deformations and is weighted by w_{SE} . The optimal β is iteratively solved using the Levenberg–Marquardt algorithm until the change in $\Omega(\beta)$ satisfies $|\Delta\Omega(\beta)| < 10^{-12}$ [23].

III. EXPERIMENT

To validate the system's feasibility, we performed a phantom study. Fig. 3 shows the overall workflow.

A. Phantom & Phantom Base Fabrication

To ensure a realistic breast shape, we took a healthy volunteer's MRI scan from the same IRB approved study in Section II-B1 to generate an inverse mold of the breast with two side extensions for applying deformations. The mold was fabricated using a 3D printer (Bambu Lab X1, Shenzhen, Guangdong, China). To ensure the phantom has similar mechanical properties as breast tissues, we followed the recipe described in [24]. Specifically, the breast phantom was constructed of 56% Ecoflex 00-10 platinum cure silicone rubber, 7% silicone rubber thinning fluid, 7% tactile mutator, Silc PigTM PMS 488 C (all from Smooth-On Inc., Macungie, PA, USA), and 30% 350 CST silicone oil (purchased from Consolidated Chemical & Solvents, Amazon.usa) based on mass, and 90% Nylon and 10% Spandex fabric wrap (purchased from Barcelonetta, Amazon.usa). Next, Silc PigTM PMS 2757 C (Smooth-On Inc., Macungie, PA, USA) was mixed with the same rubber and then used to inscribe the 26 letters onto the phantom. Then, 26 red plastic fiducial markers were embedded on the breast surface and sealed with the same rubber mixture without dye. For subsurface targets, 16 metal beads were evenly embedded into the breast phantom using an implant needle, primarily within regions where tumors are commonly found in BCS. Both plastic fiducial markers and metal beads were visible and identifiable in the CT image.

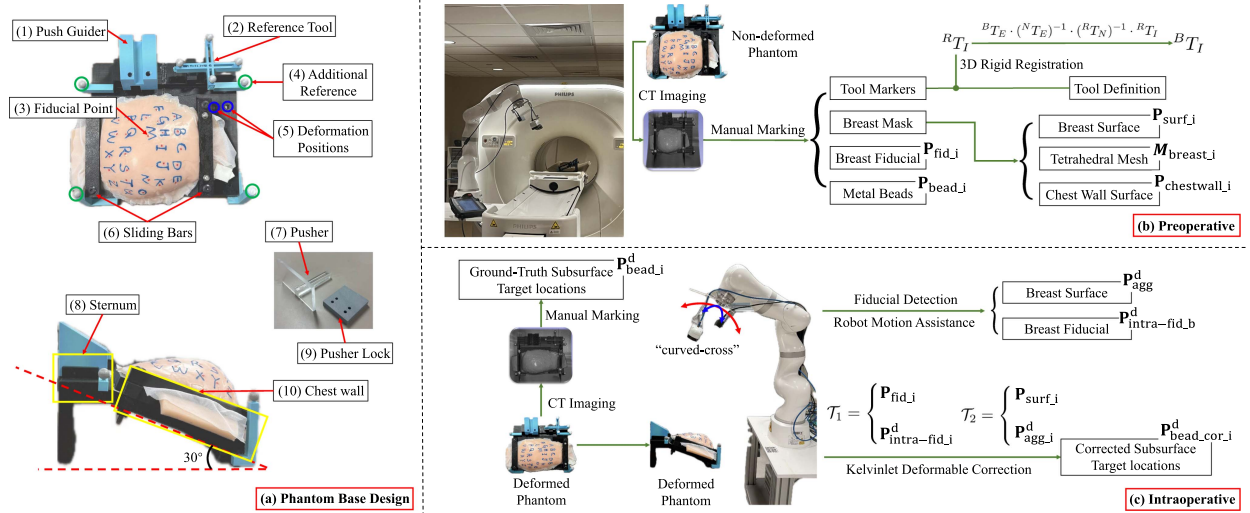


Fig. 3. Experimental Workflow. (a) Phantom base design; (b) Preoperative workflow: the undeformed phantom was positioned in the CT scanner to acquire preoperative imaging data; (c) Intraoperative workflow: for each fixed deformation state, after acquiring ground-truth imaging, the phantom was translated out of the PET/CT scanner using the motorized bed, and intraoperative data were immediately acquired using the robot.

As shown in the Fig. 3(a), the phantom base consists of two parts, representing the sternum (Fig. 3(a) (8)) and the chest wall (Fig. 3(a) (10)). A reference tool (Fig. 3(a) (2)) is fixed on the sternum, and the chest wall was designed with an inclination angle of 30° according to the volunteer's data. The sliding bars (Fig. 3(a) (6)) were designed to restrain both sides of the breast phantom and to apply stretching deformation. In the sternum region, the pusher (Fig. 3(a) (7)) was used to apply compressive deformation. Based on multi-posture MRI scans, we designed two stretching deformation states by using the two deformation positions (Fig. 3(a) (5)) to simulate arm position changes, and one composite deformation state that combined stretching and compressive deformation to represent the arm 90° extension. Other than the reference tool, four additional retro-reflective passive sphere markers (Fig. 3(a) (4)) surrounding the base were attached for obtaining ${}^R T_I$ (Section II-C1) while obtaining ground-truth subsurface target locations. All components were 3D-printed using the same machine in Section III-A.

B. Experimental Setup

We acquired repeat CT images using a Philips Vereos Time-of-Flight CT system with a slice thickness of 1 mm. To obtain preoperative data, a CT scan of the non-deformed phantom was processed in 3D Slicer (Kitware Inc, Clifton Park, NY) to annotate the surface fiducial marker positions $\mathbf{P}_{\text{fid}_i}$, subsurface target positions $\mathbf{P}_{\text{bead}_i}$, and reference tool points $\mathbf{P}_{\text{ref}_i}$. The ${}^R T_I$ was then computed (Section II-C1). The breast region was segmented to generate a binary mask, which was decimated and processed into a sparse tetrahedral mesh $\mathcal{M}_{\text{breast}_i}$. The breast surface and chest wall surface, $\mathbf{P}_{\text{surf}_i}$ and $\mathbf{P}_{\text{chestwall}_i}$, were then extracted from $\mathcal{M}_{\text{breast}_i}$. Then, $k = 45$ control points in Section II-C were generated from the $\mathbf{P}_{\text{chestwall}_i}$ using the same method [8].

To calculate subsurface target locations, we configured the phantom base to reflect a new unique deformation for each

deformation state d . As shown in Fig. 3(c), after the ground-truth CT scan was first obtained for each sequential deformation, the ground-truth subsurface targets $\mathbf{P}_{\text{bead}_i}^d$ were obtained, and the robot then performed the steps in Section II-B2 to acquire intraoperative data $\mathbf{P}_{\text{intra-fid}_b}^d$ and $\mathbf{P}_{\text{agg}}^d$. We set the parameters as $R = 0.7$ m, $\Delta\theta = 2^\circ$, $\theta_{\text{start}} = 32^\circ$, $\theta_{\text{end}} = 15^\circ$, $\phi_{\text{start}} = 5^\circ$, and $\phi_{\text{end}} = -5^\circ$ to ensure complete surveillance of breast surface. Then, $\mathbf{P}_{\text{intra-fid}_b}^d$ was manually inspected to remove false detections or correct any pair $(\mathbf{P}_{\text{intra-fid}_b,j}^d, \ell_j^d)$ where ℓ_j^d does not correspond to $\mathbf{P}_{\text{intra-fid}_b,j}^d$. Subsequently, $\mathbf{P}_{\text{intra-fid}_b}^d$ and $\mathbf{P}_{\text{agg}}^d$ (with $\eta = 6$ mm) were transformed from the robot base space to CT imaging space as $\mathbf{P}_{\text{intra-fid}_i}^d$ and $\mathbf{P}_{\text{agg}_i}^d$ by using $({}^B T_I)^{-1}$, where ${}^B T_I = {}^B T_E \cdot ({}^N T_E)^{-1} \cdot ({}^R T_N)^{-1} \cdot {}^R T_I$. Finally, the following two groups of data, $\mathcal{M}_{\text{breast}_i}$, $\mathbf{P}_{\text{bead}_i}$, and the control points were used as inputs for the deformable correction pipeline, yielding the corrected subsurface target locations $\mathbf{P}_{\text{bead_cor}_i}^d$:

$$\mathcal{T}_1 = \begin{Bmatrix} \mathbf{P}_{\text{fid}_i} \\ \mathbf{P}_{\text{intra-fid}_i}^d \end{Bmatrix}, \quad \mathcal{T}_2 = \begin{Bmatrix} \mathbf{P}_{\text{surf}_i} \\ \mathbf{P}_{\text{agg}_i}^d \end{Bmatrix}, \quad (3)$$

where $\mathcal{T}_1$ and $\mathcal{T}_2$ provide the data for the first and second terms in (2), respectively.

To obtain TREs for rigid and non-rigid registration, $\{\mathbf{P}_{\text{bead}_i}^d\}_{d=1}^3$, $\{\mathbf{P}_{\text{bead_cor}_i}^d\}_{d=1}^3$, and $\mathbf{P}_{\text{bead}_i}$ were transformed to the reference tool space as $\{\mathbf{P}_{\text{bead}_r}^d\}_{d=1}^3$, $\{\mathbf{P}_{\text{bead_cor}_r}^d\}_{d=1}^3$, and $\mathbf{P}_{\text{bead}_r}$ using ${}^R T_I$ from their corresponding CT scans. The non-rigid registration TRE is defined as the Euclidean distance $\|\mathbf{P}_{\text{bead_cor}_r,j}^d - \mathbf{P}_{\text{bead}_r,j}^d\|_2$ for each corresponding pair j , and the rigid registration TRE as $\|\mathbf{P}_{\text{bead_cor}_r,j}^d - \mathbf{P}_{\text{bead}_r,j}^d\|_2$. As noted in the clinical standard of care, subsurface targets are commonly localized intraoperatively via implanted seed-based marker techniques, in which a specialized probe tracks the subsurface seed location to assist in detecting a lesion. Motivated by this, we further performed a spatial region of

interest (sROI) analysis following [11]. Specifically, an sROI is defined as a sphere of radius r centered at a designated *seed* (denoted as s). We iteratively treated one subsurface target $\mathbf{P}_{\text{bead_i},s}^d$ from $\mathbf{P}_{\text{bead_i}}^d$ as a *seed*, augmenting $\mathbf{P}_{\text{intra-fid_i}}^d \leftarrow \{\mathbf{P}_{\text{intra-fid_i}}^d, \mathbf{P}_{\text{bead_i},s}^d\}$ and re-running the deformable correction pipeline. The sROI analysis TRE is computed in the same manner as the non-rigid registration TRE above, but only for $\mathbf{P}_{\text{bead_cor_r},j}^d \in \mathcal{A}_s^d$, where $\mathcal{A}_s^d = \{\mathbf{P}_{\text{bead_cor_r},j}^d \mid j \neq s, \|\mathbf{P}_{\text{bead_cor_r},j}^d - \mathbf{P}_{\text{bead_r},s}^d\|_2 \leq r\}$. All TRE results are summarized in Section IV.

C. System Robustness Analysis

The main sources of error may arise from multiple intermediate steps in *ReSNA*. Therefore, we performed an error propagation analysis to assess how uncertainty at each step potentially amplifies into the final TRE. Specifically, we defined a zero-mean isotropic Gaussian noise $\mathcal{N}(0, \sigma^2)$ and injected it independently into six individual error sources: (1) Lyra-Robot hand-eye translation, (2) Lyra-Robot hand-eye rotation, (3) ZED-NDI calibration, (4) surface fiducial marker localization, (5) surface point cloud acquisition, and (6) deformable correction. For each noise source, we swept σ across five levels ($\sigma \in \{0.5, 1.0, 1.5, 3.0, 5.0\}$ mm for translational sources, and $\sigma \in \{0.2, 0.5, 1.0, 1.5, 3.0\}^\circ$ for rotational sources). At each σ level, we generated 20 independent Monte Carlo samples. For error sources (1) and (2), we first generated two sets of noisy homogeneous transformations $\mathbf{E}_{A_i}$ and $\mathbf{E}_{B_i}$ for each motion pair in the hand-eye calibration. The translational component is sampled as $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2)$, while the rotational component is constructed via the Rodrigues formula $\mathbf{R}(\hat{\mathbf{k}}, \theta)$, with $\hat{\mathbf{k}}$ a uniformly random unit axis and $\theta \sim \mathcal{N}(0, \sigma_{\text{deg}}^2)$. Adding $\mathbf{E}_{A_i}$ and $\mathbf{E}_{B_i}$ into $\mathbf{A}_i \mathbf{X} = \mathbf{X} \mathbf{B}_i$ yields the non-homogeneous Sylvester equation $\mathbf{A}_i \mathbf{X}' - \mathbf{X}' \mathbf{B}_i = \mathbf{C}_i$, where $\mathbf{C}_i = \mathbf{E}_{B_i} - \mathbf{E}_{A_i}$, which is solved in the least-squares sense via the vectorized linear system $(\mathbf{I} \otimes \mathbf{A}_i - \mathbf{B}_i^\top \otimes \mathbf{I}) \text{vec}(\mathbf{X}') = \text{vec}(\mathbf{C}_i)$. The resulting noise-corrupted $\mathbf{X}'$ is reinserted into the pipeline to recompute the noisy TRE. For error sources (3) through (6), we add translational noise in the same manner to the checkerboard corner points, the surface fiducial marker positions, surface point cloud, and the control point positions of the deformable correction pipeline, respectively. To evaluate the sensitivity of *ReSNA* to surface point cloud density, we varied the decimation parameter η used for $\mathbf{P}_{\text{agg}}$ in five density levels (2mm, 4mm, 6mm, 8mm, and 10mm). The $\eta = 6$ mm was set as the reference for comparisons. Both analyses were conducted across all deformation states and the results are summarized in Section IV.

IV. RESULTS

A. ReSNA Realization Specifications

For the hand-eye calibration, the rotation and translation root mean square errors (RMSE) were 0.204° and 2.31 mm. For the Lyra-ZED 2i calibration, the RMSE was 1.39 mm. For Section II-B1, three models were trained and evaluated on a workstation equipped with an NVIDIA RTX 4090 GPU, an AMD Ryzen 9 9950X CPU, and 128 GB of RAM with the following configurations and results:

TABLE I
OVERALL RIGID AND NON-RIGID REGISTRATION RESULTS. SUBSURFACE TARGETS TREs REPORTED IN MM. SFMS: SURFACE FIDUCIAL MARKERS; D: DETECTED; C: CORRECTED; V: VISIBLE

	Rigid Reg.	Non-rigid Reg.	Valid SFMs
	Avg $\pm$ std	Avg $\pm$ std	D+C V
State 1	8.87 ± 4.43	4.48 ± 1.57	21+1 26
State 2	15.44 ± 8.97	2.78 ± 0.87	20+1 26
State 3	9.75 ± 5.14	2.82 ± 1.03	17+2 25
Average	11.32 ± 7.01	3.36 ± 1.41	19-21

TABLE II
sROI ANALYSIS RESULTS. *ReSNA* SMALL-REGION LOCALIZATION PERFORMANCE REPORTED IN MM

	$r = \infty$	$r = 30$ mm	
	Non-rigid Reg.	Non-rigid Reg.	#sROI spheres
	Avg $\pm$ std	Avg $\pm$ std	Count
State 1	3.16 ± 1.38	2.35 ± 1.24	16
State 2	3.04 ± 1.45	2.93 ± 1.55	16
State 3	3.66 ± 1.85	2.91 ± 1.70	16
Average	3.29 ± 1.59	2.74 ± 1.53	16

- **LocNet**: optimized using a mean-squared-error (MSE) loss, Adam optimizer with a learning rate of 1×10^{-5} , and a batch size of 40 for 200 epochs, yielding a test MSE of 0.0171.
- **RF-DETR**: fine-tuned for 100 epochs with a batch size of 16, a learning rate of 1×10^{-4} , and the default configuration [19] for others, and achieved 51.3% mAP@50:95 and 69.2% mAP@50.
- **Pose Anything**: trained using same configuration [20] and achieved fiducial point detection accuracies of 81.82% and 89.09% under pixel distance thresholds of 5 and 10 pixels, respectively.

The system operated on the workstation with the ZED 2i frame rate set to 30 Hz. The robot motion frequency was set to 5 seconds between consecutive poses, resulting in a total execution time of 2–3 minutes for one full *curved-cross* trajectory. Upon visual supervision of $\{\mathbf{P}_{\text{intra-fid_b}}^d\}_{d=1}^3$, one mislabeled fiducial marker was corrected in $d = \{1, 2\}$, whereas in $d = 3$, two mislabeled fiducial markers were corrected and one mislocalized fiducial marker was excluded. Each deformable correction took, on average, 10 seconds to obtain $\mathbf{P}_{\text{bead_cor_i}}^d$.

B. Targeting Study and Uncertainty Analysis

The overall rigid and non-rigid TRE results across the three deformation states are reported in Table I, demonstrating a substantial reduction from the rigid TRE (11.3 ± 7.0 mm) to the non-rigid TRE (3.4 ± 1.4 mm). As a representative example, the subsurface target spatial distribution for the third (composite) deformation state is visualized in Fig. 4, where a clear improvement after deformable correction can be observed (notice how the corrected (red square) is more closely aligned with the ground truth (green circle) in comparison to the rigid (blue triangle)). The sROI analysis, summarized in Table II ($|\mathcal{A}_s^d| \in [1, 3]$ for

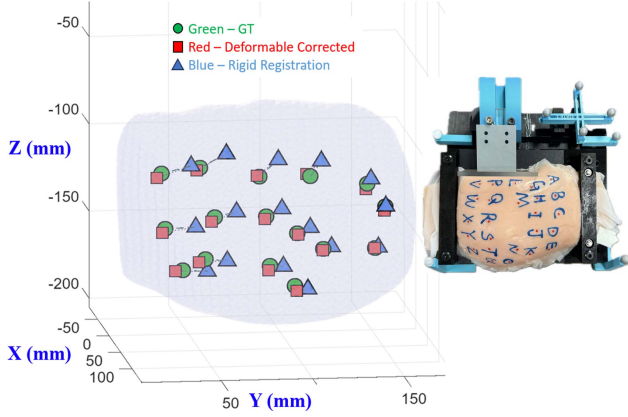


Fig. 4. Deformation State 3 Subsurface Targets Plot. The cyan rendering represents the deformable corrected breast mesh from $\mathcal{M}_{\text{breast}_i}$ for this state.

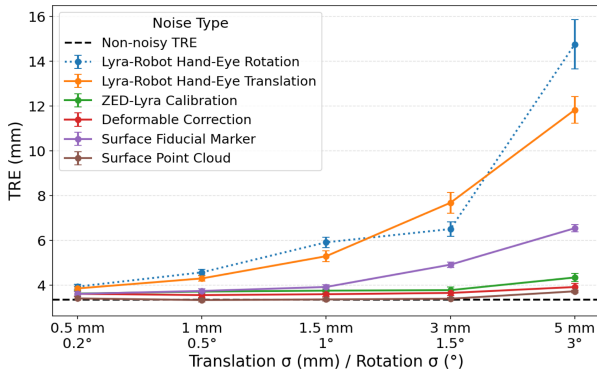


Fig. 5. Error Analysis Plot. The NDI hand-eye rotation uses rotational noise σ ($^\circ$), whereas the others use translational noise σ (mm).

TABLE III
DENSITY LEVEL TEST. EFFECT OF DENSITY LEVEL (η) CHOICE FOR $\mathbf{P}_{\text{AGG}}$ ON TRE

η (mm)	TRE (mm)	p-value	Significance
2	3.26 ± 1.37	0.074	ns
4	3.44 ± 1.35	0.183	ns
6	3.36 ± 1.41	—	reference
8	3.24 ± 1.18	0.372	ns
10	3.36 ± 1.39	0.518	ns

$r = \infty$ and $r = 30$ mm), shows a further reduction in TRE to 2.7 ± 1.5 mm with the inclusion of a single additional subsurface *seed*. The system robustness analysis is presented in Fig. 5 and Table III. All statistical comparisons were conducted using Wilcoxon signed-rank tests at a significance level of $\kappa = 0.05$, with the detailed outcomes discussed in Section V.

V. DISCUSSION

From Table II, the proposed *ReSNA* with non-rigid registration significantly reduced the TRE compared to rigid registration, with an improvement of **70.3%** ($p < 0.001$). This is likely due to several factors: (1) automated robotic assistant optimizes intraoperative data acquisition, (2) the enhanced deformable correction provides a better initial alignment, and (3) our phantom

experiments with simplistic excitation are more closely mimicked by our elastic model than the in vivo counterpart in [11]. Our surface fiducial marker detection and tracking pipeline in Section II-B1 automatically detected an average of 19 fiducials and reached 21 with minimal corrections, substantially reducing manual burden while maintaining comparable accuracy [18]. The sROI analysis uses the subsurface information from a single seed to assist in driving the alignment and error is assessed throughout the volume of subsurface targets ($r = \infty$), and a subset within 30 mm ($r = 30$ mm). Overall, while modest, the improvement in TRE for the $r = 30$ mm instance was statistically significant over the results in Table I (non-rigid) and Table II ($r = \infty$), with $p = 0.0179$ and $p = 0.0014$, respectively. These findings suggest that *ReSNA* can achieve more precise localization by integrating seed-based techniques that are widely adopted in clinical practice. From a clinical standpoint, 77–87% of breast lesion resection volumes have radii smaller than 10 mm [25], and a resection margin of 20 mm radius is typically recommended [26]. Under the 2 mm minimum margin standard [27], the achieved TREs of 3.4 mm without a *seed* and 2.7 mm with a *seed* are clinically sufficient to ensure the tumor remains within the resection volume. Regarding system robustness, the error propagation analysis in Fig. 5 identifies hand-eye calibration as the dominant source of sensitivity and intraoperative surface fiducial marker localization as a secondary contributor; both are controllable through rigorous hand-eye calibration protocols and manual inspection of the acquired markers, respectively. The density sensitivity analysis in Table III further demonstrates that TRE remains stable across all tested decimation levels (all $p > 0.05$), confirming *ReSNA*'s compatibility with varying point cloud resolution and computational capability. Collectively, these results indicate that *ReSNA* maintains robust performance across its primary components, supporting its adaptability to diverse operating environments and cost-effective substitution of components with functionally equivalent alternatives.

While this study presents an early stage *ReSNA*, future work will focus on in vivo evaluation, extending nonrigid alignment approaches to enable resection tracking for margin navigation, developing surgeon-interfacing guidance to enable intraoperative navigation, creating more intelligent robotic surveillance strategies, incorporating implicit correspondence [28] into surface marker design, and integrating physics-aware learned surrogates [29] to accelerate intraoperative computation.

VI. CONCLUSION

We presented *ReSNA*, a novel breast surveillance system for BCS that employs an “eyes-in-hand” robotic platform. Unlike most existing robotic surgical systems serving as a tool extension to the surgeon, *ReSNA* is being designed to work as an independent agent providing additional automated eyes measuring the patient’s anatomy. More specifically, *ReSNA* mobilizes a diversified set of computer-vision resources and dynamic robot positioning to achieve precise texture-based correspondence and dense surface geometry acquisition. These capabilities are then further coupled with deformable correction that enables

accurate intraoperative tumor localization while preserving surgical workflow. Although additional development and clinical validation are required, *ReSNA* demonstrates its strong potential for translation to a BCS image-guidance system (BCS-IGS) as well as other soft-tissue IGS.

APPENDIX NETWORK DETAILS

LocNet is a heatmap regression network built on an ImageNet-pretrained [30] ResNet-18 backbone [31]. The backbone is truncated before the global average pooling layer and followed by a 3×3 convolution, two transposed-convolutional layers (stride 2), and a final 1×1 convolution with sigmoid activation to produce a single-channel likelihood map indicating the probability of breast surface fiducial markers presence. For the Pose Anything model training, we randomly selected breast images from the training, validation, and testing sets to extract 537, 177, and 165 single-letter images across 26 categories ('A'–'Z') correspondingly, yielding a total of 879 images. Each image was then annotated with '4'–'8' pose keypoints, designating the fiducial point as the first keypoint. The same data preprocessing [20] was adopted.

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